Deep Computer Vision Models for Speech Emotion Recognition

Kristjan Kostanjšek (63170154)

Abstract

This report explores a computer vision-based approach to Speech Emotion Recognition (SER) using Mel spectrograms as image inputs. Evaluated on the challenging CREMA-D dataset, the study compares ResNet18, ConvNeXt-Tiny and DINOv2. Through techniques like transfer learning, data augmentation, and multi-task learning (MTL), the best CNN model achieved 67.69% accuracy, significantly exceeding human-level performance of ∼40%. While vision-based models proved promising, they were ultimately outperformed by a wav2vec 2.0 baseline (71.46%), highlighting the advantages of self-supervised audio pretraining over domain-shifted visual representations.

1 Introduction

Speech emotion recognition (SER) is the task of automatically identifying human emotional states from speech signals. SER plays an important role in advancing Human-Computer Interaction (HCI), enabling systems to respond not only to what users say, but also to how they say it. The ability to detect emotions from speech has a wide range of real-world applications, particularly in domains where understanding human affection can improve interaction quality, safety and decision-making. For example, emotion-aware virtual assistants can adapt their responses based on the user's mood, customer service systems can detect frustration or anger to trigger appropriate interventions and healthcare applications can monitor patients' emotional states for early signs of distress or mental health issues.

Traditional approaches to SER relied heavily on classical machine learning models such as Hidden Markov Models (HMMs), Gaussian Mixture Models (GMMs), and Support Vector Machines (SVMs). While these methods achieved moderate success, their performance was somewhat limited. The shift to deep learning has significantly improved SER performance by enabling models to learn hierarchical representations directly from data. Deep learning approaches to SER rely on everything from CNNs, to RNNs and LSTMs and recently transformer-based models [1].

In my project I decided to train computer vision models (CNNs and vision transformer-based architectures) to approach SER, using Mel spectrograms as input, because they provide a visual representation of frequency content over time. This enables the problem to be treated as an image classification task, allowing the use of well-established architectures form the computer vision domain. The models explored in this work include ResNet, ConvNeXt and DINOv2, representing both convolutional and transformer-based approaches. While the focus remains on the aforementioned models, I will also experiment with wav2vec 2.0, which is currently regarded as one of the strongest approaches for speech-based tasks. This is intended primarily as a point of comparison, to better understand how the proposed models perform relative to a state-of-the-art method.

I will conduct the experiments on the CREMA-D dataset, which is relatively small and challenging, with human-level accuracy reported at approximately 40%. Due to the limited dataset size and task difficulty, particular attention will be given to mitigating overfitting through the use of smaller network architectures, regularization techniques, and data augmentation strategies. Additionally, the impact of transfer learning will be investigated by leveraging models pretrained on ImageNet, with a focus on understanding whether the domain gap between natural images and spectrograms benefits or hinders performance.

2 Methods

This section provides an overview of the dataset used in this work, along with the models and overall experimental pipeline. It also outlines the various techniques explored to improve model performance.

2.1 Dataset

I initailly planned to use the RAVDESS Emotional Speech Audio dataset for this project. However I will instead use CREMA-D dataset [3], as RAVDESS proved to be too limited in size for effectively training even smaller depp learning architectures.

Although CREMA-D also includes video recordings, I will focus on audio-only dataset. The dataset contains 7442 recordings from 91 actors of different races and ethnicities. Actors spoke a selection of 12 sentences, presented using six different emotions (Anger, Disgust, Fear, Happy, Neutral, and

Sad) and four different emotion levels (Low, Medium, High, and Unspecified).

The distribution of emotion classes in the CREMA-D dataset is shown in Figure 1. The classes are relatively well balanced, with the exception of the Neutral class, which is slightly underrepresented. However, I do not expect this imbalance to significantly impact the results.

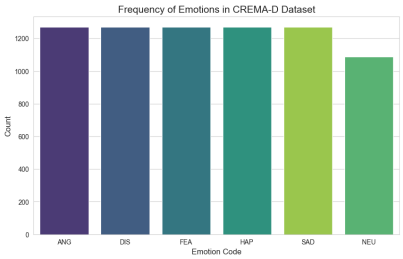

Figure 1: Frequency of emotion classes in the CREMA-D dataset.

Despite being larger than RAVDESS, CREMA-D is still relatively small for training deep learning models, with approximately 1200 samples per class. Because of this, I will focus on using smaller models that are less prone to overfitting, and I will experiment with different regularization and data augmentation strategies to improve generalization.

Another important aspect of CREMA-D is that it is considered a relatively challenging dataset. Human performance on audio-only emotion recognition reaches only 40.9%, while existing deep learning approaches report accuracies of approximately 68% [4]. I include this as a point of reference to set realistic expectations for model performance on this task.

To ensure the model learns to generalize across different voices rather than simply memorizing individual vocal characteristics, I implemented a speaker-independent split for the training, validation, and testing sets. Ideally, a completely unbiased evaluation would require splitting the data by both actors and sentences. However, this is unfeasible with the CREMA-D dataset because every actor performs the exact same 12 sentences. To achieve a split that is independent in both dimensions, we would be forced to discard a significant portion of the recordings, which is not an option given the already limited size of the dataset.

2.2 Models Used

As mentioned previously, I will focus on smaller model variants due to the limited size of the dataset and the associated risk of overfitting.

The first model I will use is **ResNet** [5], a canonical CNN architecture and the winner of the ImageNet 2015 competition. It is well known for introducing residual connections through skip connections, which enable the training of deeper networks. I chose this model because of its simplicity and robustness, while still offering competitive performance even by modern standards. In this project, I will use ResNet18, which has approximately 11.7M trainable parameters.

The second model is **ConvNeXt** [6], a more modern CNN architecture that revisits and improves traditional convolutional designs by incorporating several ideas inspired by transformer-based models. I will use the ConvNeXt-Tiny variant, which has approximately 28.6M parameters.

The third model I will use is **DINOv2** [7], which is based on a vision transformer architecture and trained using a self-supervised learning approach. It is pretrained on a large-scale unlabeled image dataset, allowing it to learn rich and generalizable visual representations. In this project, I will use the DINOv2-Small model (ViT-S/14 backbone), which has approximately 21M parameters.

Finally, I will also include **wav2vec 2.0** [2] as a reference model. Unlike the previous models, which operate on spectrograms as images, wav2vec 2.0 is designed specifically for raw audio and uses self-supervised pretraining to learn meaningful speech representations from large amounts of unlabeled data. Because of this, it is known to achieve strong performance even when only limited labeled data is available, making it a suitable upper baseline for comparison. However, I will not focus on improving this model, as the primary goal of this work is to explore and optimize the vision-based approaches.

2.3 Pipeline

In this subsection, I will briefly describe the overall processing pipeline used in this project.

First I take the raw `.wav` audio files and standardize their length by either padding or truncating them to exactly 3 seconds. I then resample all audio to 16 kHz to ensure consistency across samples. From the processed audio I compute Mel spectrograms using 128 Mel frequency bands and normalize their values to the range between 0 and 1. Next I apply various data augmentation techniques directly to the Mel spectrograms in order to improve model robustness. The augmented spectrograms are then used as inputs to the selected models. Because the models expect 3-channel input we modify the first layer to expect only 1 channel in case of CNNs, and in case of DINOv2 we modify the spectrogram, turning it into a 3-channel image. Finally the feature representations produced by the models is passed to a small multi-layer perceptron (MLP), which performs the final emotion classification using the Cross Entropy loss function.

2.4 Regularization and Overfitting Mitigation

As discussed earlier, the CREMA-D dataset is relatively small, which led to a high risk of overfitting. To address this, I employed several strategies aimed at improving generalization, which are summarized below:

- **Regularization:** I used multiple regularization techniques to stabilize training. Dropout was applied in the final MLP head, where neurons were randomly deactivated during training to reduce co-adaptation. I also used weight decay, which encourages smaller weight values by nudging them towards zero. Additionally, I applied label smoothing, where a small portion of probability mass was distributed across incorrect classes (e.g. assigning 0.9 to the correct class and 0.1 to the others), preventing the model from becoming overly confident.
- **Data augmentation:** I applied augmentation directly to the Mel spectrograms to increase input variability. This

included adding a small amount of Gaussian noise, as well as time and frequency masking. These masking techniques randomly removed consecutive time steps or frequency bands, forcing the model to rely on surrounding contextual information rather than overfitting to specific patterns. An example of resulting data augmentation is showin in Figure 2.

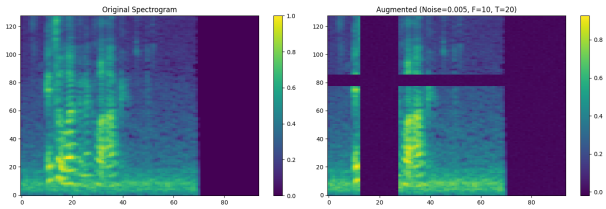

Figure 2: Original spectrogram (left) and results of data augmentation (right).

- **Learning rate scheduling:** I used a learning rate scheduler to allow for more controlled optimization and improved convergence. Specifically, I monitored validation loss and reduced the learning rate by a factor of 0.5 if no improvement was observed for 2 consecutive epochs. This helped the model make finer updates during later stages of training.
- **Pretrained models (transfer learning):** I also explored transfer learning as an additional strategy. The CNN-based models (ResNet and ConvNeXt) were pretrained on ImageNet, while DINOv2 was pretrained on the LVD-142M dataset, both consisting of natural images. This introduces a potential domain gap, as the target inputs in this project are spectrograms rather than natural images. I was therefore interested in evaluating whether this pretrained knowledge would still be beneficial or potentially hinder performance. This can be contrasted with wav2vec 2.0, which is pretrained directly on raw audio, making it more naturally aligned with the target task.

2.5 Multi-Task Learning

Another approach I explored was multi-task learning.

In addition to emotion labels, the CREMA-D dataset also provides annotations for emotion intensity (Low, Medium, High or Unspecified). Intuitively, samples with lower intensity can be more ambiguous and harder to classify correctly. Based on this I hypothesized, that incorporating intensity as an auxiliary task could encourage the model to learn richer and more discriminative feature representations. In particular, this additional supervision may help the model better distinguish between subtle emotional expressions.

To implement this, I introduced an additional output head for emotion intensity prediction alongside the main emotion classification head. The primary task (emotion classification) used Cross Entropy loss, while the auxiliary task (intensity prediction) was formulated as a regression problem using SmoothL1 loss. The total loss was computed as:

$$\mathcal{L}_{total} = \mathcal{L}_{emotion} + \alpha \cdot \mathcal{L}_{intensity}$$

where α is a weighting hyperparameter used to balance the contribution of the auxiliary task. I set $\alpha = 0.3$ to ensure that the intensity loss doesn't dominate the main classification objective.

One limitation of this approach is that a significant portion of the CREMA-D samples have unspecified intensity labels. Instead of discarding these samples, I mapped the "unspecified" label to the mean value of the intensity range for the auxiliary regression task. This allowed the model to leverage all available data while providing a neutral target value for samples where a specific intensity was not annotated. While this is not a perfectly clean formulation of the problem, it allowed me to make use of all available data.

2.6 Intermediate Feature Extraction

The final improvement I explored was intermediate feature extraction.

Instead of relying solely on the final feature vector produced by the backbone, I also extracted features from intermediate layers. To ensure compatibility, these intermediate feature maps were reduced using Global Average Pooling, resulting in a single value per channel. The pooled intermediate features were then concatenated with the final feature vector and passed to the MLP classifier.

The motivation behind this approach is that different layers capture different levels of abstraction. Earlier layers tend to focus on low-level patterns such as edges and textures, while deeper layers capture more task-specific and high-level representations. Incorporating intermediate features therefore allows the model to leverage information across multiple representation levels. This is particularly relevant in this setting, as the models were pretrained on natural images and then applied to spectrograms, which represent a different domain. In such cases, earlier features may still remain useful, while deeper features may be less well aligned with the target task.

However, this approach also increases the dimensionality of the input to the MLP. For example, in the case of ResNet18, the feature vector size increased from 512 to 960. This nearly doubles the number of parameters in the classifier head, which can exacerbate overfitting, especially given the limited size of the dataset.

3 Results and Analysis

In this section, the experimental results are presented and analyzed.

3.1 Training Dynamics and Progressive Improvements on CNN Backbones

Figures 3 and 4 show the training and validation accuracy curves for ResNet18 and ConvNeXt-Tiny, respectively. Each figure contains four configurations: (1) a baseline model trained from scratch, (2) a pretrained model, (3) a pretrained model with regularization and data augmentation, and (4) the final configuration, which additionally includes a learning rate scheduler.

Several consistent trends can be observed across both architectures. The baseline ResNet18 exhibited strong overfitting, where training accuracy increased rapidly while validation accuracy remained largely unchanged. In contrast,

the baseline ConvNeXt-Tiny showed slower but more stable learning, with less obvious overfitting, although overall performance improved only gradually.

With the introduction of transfer learning, both models achieved a significant boost in performance. This suggests that, despite the domain gap between natural images and spectrograms, pretrained features still provided a strong initialization. However using pretrained models also introduced heavier overfitting, as the models were able to fit the training data much more effectively.

Adding data augmentation and regularization helped mitigate overfitting to some extent. The gap between training and validation performance decreased, and the models began to show more meaningful learning behavior. However, the validation accuracy still exhibited noticeable fluctuations, indicating unstable optimization.

This instability was largely addressed by introducing the learning rate scheduler, which resulted in smoother and more consistent validation curves. Across both models, validation accuracy typically increased rapidly in the early stages of training and then stagnated at around 60%.

The final configurations (combining pretraining, regularization, data augmentation, and learning rate scheduling) achieved test accuracies of **64.50%** for ResNet18 and **64.13%** for ConvNeXt-Tiny (best results over 10 runs). The performance of the two models is therefore very similar. The small difference in accuracy can likely be attributed to variance across runs, although it is also possible that the smaller parameter count of ResNet18 makes it slightly better suited for this relatively limited dataset.

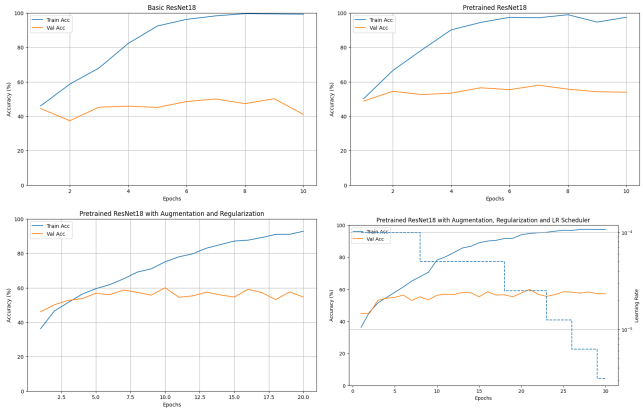

Figure 3: ResNet18 train and validation accuracy during training for basic model (top left), pretrained model (top right), pretrained model trained with regularization and data augmentation (bottom left) and pretrained model trained with regularization, data augmentation and LR scheduler (bottom right).

3.2 DINOv2 Results with Different Backbone Freezing Strategies

For the DINOv2 model, I used pretrained weights from the start, as leveraging large-scale self-supervised pretraining is a core aspect of this architecture. In addition, I applied the same training strategies as before, including regularization, data augmentation and a learning rate scheduler.

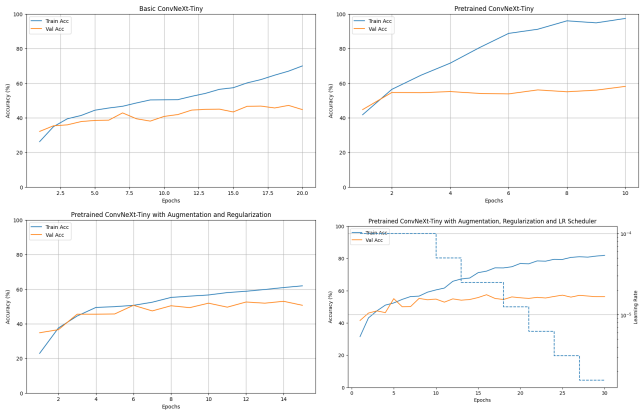

Figure 4: ConvNeXt-Tiny train and validation accuracy during training for basic model (top left), pretrained model (top right), pretrained model trained with regularization and data augmentation (bottom left) and pretrained model trained with regularization, data augmentation and LR scheduler (bottom right).

A key part of the experiments was exploring different backbone freezing strategies during training. Specifically, I tested freezing the entire backbone, fine-tuning only the last block, fine-tuning the last three blocks, and fine-tuning the entire backbone. The results of these experiments are summarized in Table 1, where I report the best accuracy over 10 runs for each configuration.

Table 1: DINOv2 model accuracy comparison

Model Configuration	**Accuracy (%)**
freeze whole backbone	43.24
finetune last block	56.88
finetune last 3 blocks	57.13
finetune the whole backbone	44.84

The results show that both extremes (freezing the entire backbone and fine-tuning it fully) led to the worst performance. Freezing the backbone limits the model to only training the final MLP head, forcing it to rely entirely on features learned from natural images, which do not fully align with spectrogram inputs. On the other hand, fully fine-tuning the model introduces a large number of trainable parameters, which is problematic given the limited size of the dataset.

The best performance was achieved by fine-tuning only the last few blocks of the backbone. This provides a balance between adaptability and generalization, allowing the model to slightly adjust higher-level representations while keeping most pretrained features intact.

Despite this, the overall performance of DINOv2 remained noticeably below that of the CNN-based models. One possible explanation is that transformer-based architectures typically require larger amounts of data to fully realize their potential. Based on these results, I decided to focus subsequent experiments on CNN-based models and did not continue further exploration with DINOv2.

3.3 Ablation Study on MTL and IFE

In this section, I analyzed the impact of Multi-Task Learning (MTL) and Intermediate Feature Extraction (IFE) on the performance of the CNN-based models. The results for different configurations are summarized in Table 2, where I report the best accuracy obtained over 10 runs for each setting.

Table 2: Ablation study: Accuracy (%) of ResNet18 and ConvNeXt-Tiny with MTL and IFE components

Configuration	ResNet18	ConvNeXt-Tiny
No MTL/IFE	64.50	64.13
MTL	67.57	67.69
MTL + IFE	65.72	66.95

The results show a clear improvement when MTL is introduced. Both ResNet18 and ConvNeXt-Tiny benefited from the auxiliary intensity prediction task, with performance increasing by approximately 3%. This suggests that, despite the limitations of the intensity labels (the presence of unspecified values), the additional supervision helped the models learn more robust and discriminative feature representations.

In contrast, the addition of IFE did not lead to further improvements. While the idea of incorporating features from multiple levels of abstraction is intuitively appealing, the results indicate a slight decrease in performance for both models. A likely explanation is that the increased feature dimensionality introduced by IFE led to a higher overfitting, which outweighs its potential benefits in this low-data setting.

3.4 Confusion Matrix Analysis

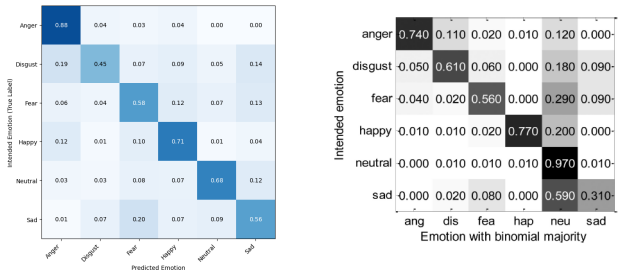

Figure 5: Confusion matrices for my final configuration of ResNet18 with MTL (left) and human accuracy (right) from CREMA-D paper [3].

The confusion matrices for the final ResNet18 model with MTL and the human performance reported in the CREMA-D paper [3] are shown in Figure 5.

It is important to note that these two matrices are not directly comparable. The CREMA-D confusion matrix includes only samples with clear annotator agreement (i.e. majority consensus), and it aggregates results across multiple modalities (audio-only, visual-only, and audio-visual). In contrast, my results are based solely on audio data and include all samples. Therefore, the comparison is only intended to provide qualitative insights rather than a strict evaluation.

Despite these differences, several interesting patterns can be observed. One notable difference is the strong bias toward the Neutral class in human predictions, which likely reflects uncertainty, i.e. when annotators are unsure, they tend to default to a neutral label.

At the same time, there are clear similarities between the two matrices. Anger appears to be among the most easily recognizable emotion in both cases, while Disgust, Fear, and Sadness are more frequently confused with one another. In particular, Sadness is often misclassified as Fear, indicating that these emotions share overlapping acoustic characteristics.

From the model's perspective, Disgust is the most challenging class, with only around 45% correct predictions. Misclassifications for this class are relatively evenly distributed across the remaining emotions, suggesting that the model struggles to learn distinctive features for it. Overall, these patterns highlight that certain confusions are inherent to the task itself, rather than being purely model-specific.

3.5 wav2vec 2.0 Baseline

Using wav2vec 2.0, I achieved a test accuracy of **71.46%** using only the pretrained model, without applying additional techniques such as regularization, data augmentation, or further architectural modifications. This result outperformed the best CNN-based model by approximately 4%.

One likely reason for this performance gap is the smaller domain mismatch. wav2vec 2.0 is pretrained directly on raw audio, which closely matches the input modality of the target task. In contrast, the CNN-based models were pretrained on natural images and then applied to spectrogram representations, introducing a larger domain shift.

Another important factor is the feature extraction process. In my approach, Mel spectrograms were computed manually as a preprocessing step. In wav2vec 2.0, however, the feature extraction pipeline is learned directly from raw audio, allowing the model to optimize representations specifically for the task.

Despite this, the CNN-based models achieved performance relatively close to wav2vec 2.0, which suggests that the vision-based approach remains competitive. At the same time, it is important to note that wav2vec 2.0 was not extensively tuned in this work. Additional improvements, such as fine-tuning strategies or regularization, could potentially lead to even higher performance.

4 Conclusion

In this work, I explored a computer vision-based approach to speech emotion recognition by treating Mel spectrograms as image inputs. Using CNN-based architectures and various improvements, I achieved a best accuracy of **67.69%**. Considering the inherent difficulty of the CREMA-D dataset and the relatively low human-level performance, this result can be considered satisfactory.

Among the evaluated models, CNN-based approaches outperformed the vision transformer-based model. However, this should not be interpreted as a general conclusion about model superiority, but rather as a consequence of the limited data

regime, where more parameter-efficient architectures are better suited.

Another key finding is that transfer learning provided substantial benefits, even when there was a clear domain gap between pretraining data (natural images) and the target task (spectrograms). Additionally, incorporating multi-task learning further improved performance, demonstrating that auxiliary supervision can be useful even when labels are imperfect.

Despite these results, the wav2vec 2.0 baseline achieved higher performance with minimal effort, reaching 71.46% accuracy. This highlights the advantage of models pretrained directly on raw audio using self-supervised learning, as they avoid the need for manual feature extraction and benefit from a smaller domain gap.

Overall, while the vision-based approach proved to be competitive, the results suggest that the future of speech emotion recognition will likely rely on methods such as wav2vec 2.0, which leverage large-scale self-supervised learning on audio data. A promising direction for future work would be to incorporate multimodal information, such as combining audio and video, to further improve performance.

References

[1] Babak Joze Abbaschian et al. "Deep Learning Techniques for Speech Emotion Recognition, from Databases to Models". In: *Sensors* 21.4 (2021). ISSN: 1424-8220. DOI: 10.3390/s21041249. URL: https://www.mdpi.com/1424-8220/21/4/1249.

[2] Alexei Baevski et al. "wav2vec 2.0: a framework for self-supervised learning of speech representations". In: *Proceedings of the 34th International Conference on Neural Information Processing Systems*. NIPS '20. Vancouver, BC, Canada: Curran Associates Inc., 2020. ISBN: 9781713829546.

[3] Houwei Cao et al. "CREMA-D: Crowd-sourced emotional multimodal actors dataset". In: *IEEE transactions on affective computing* 5 (Oct. 2014), pp. 377–390. DOI: 10.1109/TAFFC.2014.2336244.

[4] Francesco Dal Rì et al. "Speech Emotion Recognition and Deep Learning: An Extensive Validation Using Convolutional Neural Networks". In: *IEEE Access* PP (Oct. 2023), pp. 1–1. DOI: 10.1109/ACCESS.2023.3326071.

[5] Kaiming He et al. *Deep Residual Learning for Image Recognition*. 2015. arXiv: 1512.03385 [cs.CV]. URL: https://arxiv.org/abs/1512.03385.

[6] Zhuang Liu et al. "A ConvNet for the 2020s". In: *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2022, pp. 11966–11976. DOI: 10.1109/CVPR52688.2022.01167.

[7] Maxime Oquab et al. *DINOv2: Learning Robust Visual Features without Supervision*. 2024. arXiv: 2304.07193 [cs.CV]. URL: https://arxiv.org/abs/2304.07193.